Focus for this lecture

Acquire Raw Data → Prepare / Clean Data, Visualization → Feature Engineering → Pick Model and Hyper-params for Task → Model Training / Optimization → Evaluate Model Performance → Deploy Model

- Performance Evaluation Metrics
  - Accuracy Metrics
  - Precision and Recall
  - F-measure (F1 Score)
- ROC and PR curves

Accuracy, Precision And Recall

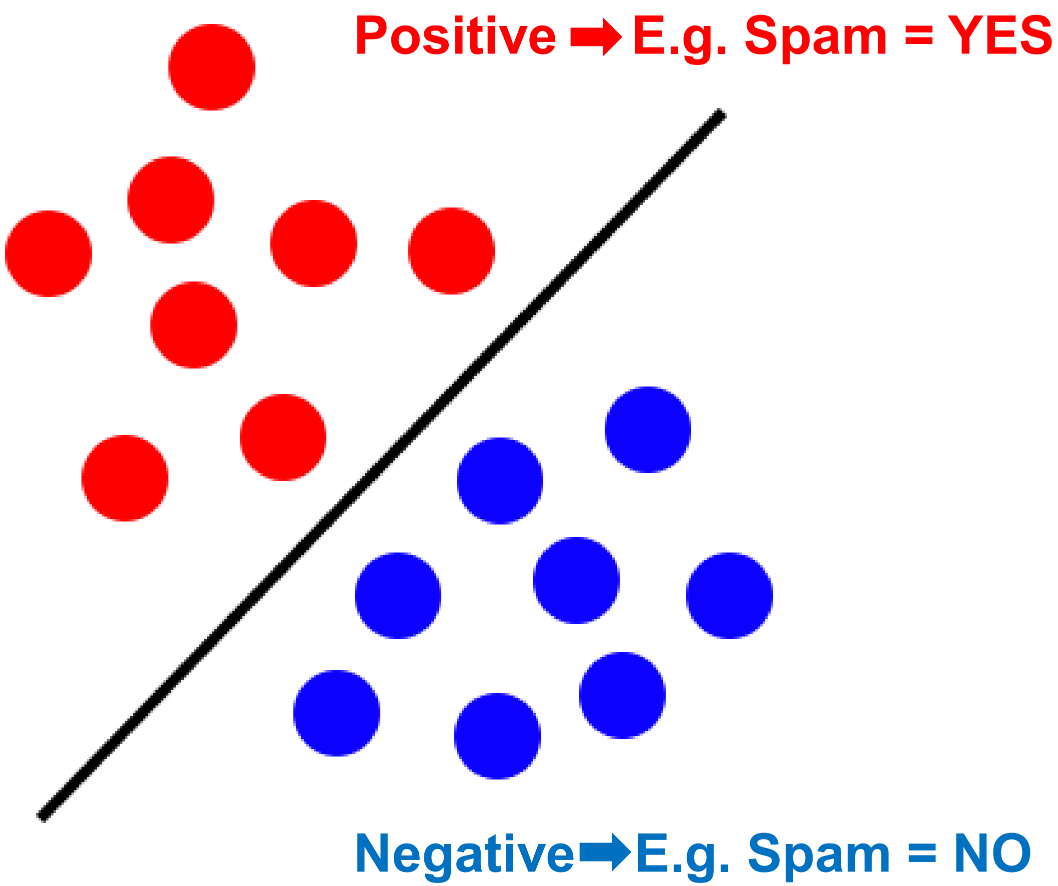
Performance Evaluation Metrics

Acquire Raw Data → Prepare / Clean Data, Visualization → Feature Engineering → Pick Model and Hyper-params for Task → Model Training / Optimization → Evaluate Model Performance → Deploy Model

- Accuracy Metrics

Accuracy Metrics

Revisiting Binary case...



Revisiting Binary case...

$$Accuracy = \frac{(100+50)}{165} = 0.91$$

$$Misclassification = \frac{(10+5)}{165} = 0.09$$

$$TruePositiveRate(TP) = \frac{(100)}{105} = 0.95$$

$$FalsePositiveRate(FP) = \frac{(10)}{60} = 0.17$$

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

Revisiting Binary case...

$$TrueNegativeRate(TN) = \frac{(50)}{60} = 0.833$$

$$FalseNegativeRate(FN) = \frac{(5)}{105} = 0.048$$

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

Key accuracy measures and terminologies

- Classification Error = $\dfrac{errors}{total}$

$$= \frac{FP + FN}{TP + TN + FP + FN}$$

- Accuracy = 1 - Error = $\dfrac{correct}{total}$

$$= \frac{TP + TN}{TP + TN + FP + FN}$$

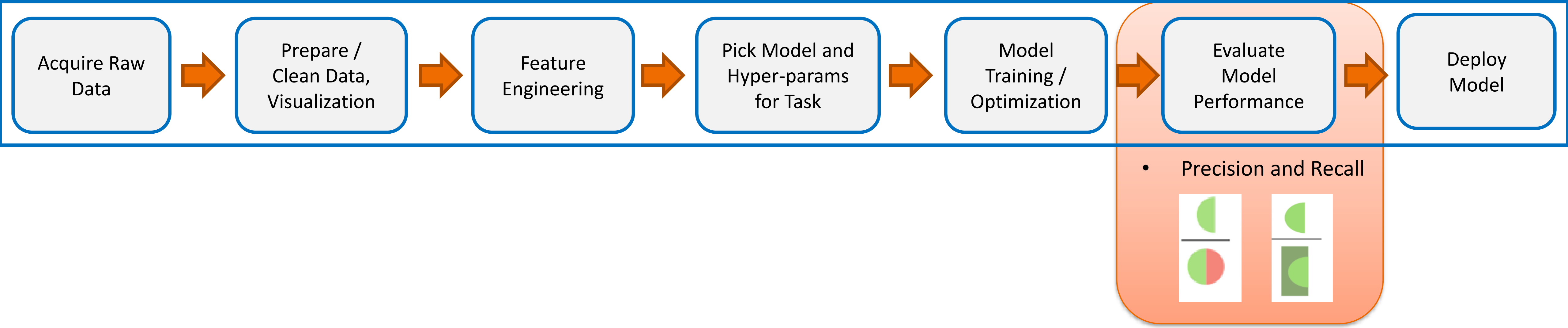
n=165	**Predicted: NO**	**Predicted: YES**	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

Revisiting scenarios where metrics are appropriate

- When you do cancer screening what do you care?
  - High TP and Low FN

- When you classify between “apple” and “orange”
  - High Accuracy

- Automatic Firing on detecting a violation.
  - Very low FP



Precision and Recall

Precision and Recall

$$Precision = \frac{TP}{(TP + FP)}$$

$$Recall = \frac{TP}{(TP + FN)}$$

Precision and Recall – examples

- **Cancer-Prediction System**
- **Pool of 100 patients' data**
- 3 patients are selected for chemotherapy ; Rest (100-3=97) are declared healthy !
- 1 year later …
- 1 of them did not actually have cancer ! (FP)
- Precision = 2/(2+1) = 67%
- 3 from the 97 healthy declared ones have cancer (FN)
- Recall = 2/(2+3) = 40%
- Accuracy = (94+2)/100 = 96%

Key accuracy measures and terminologies

- n = # of patients who underwent a new cancer screening test
- Recall = Probability of the test result being +ve given that only cancer patients are examined

$$\frac{TP}{TP + FN}$$

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

- Precision = Probability of actually having cancer given the test result is +ve

$$\frac{TP}{TP + FP}$$

Precision and Recall – examples

- A system which needs to launch a missile at a terrorist hideout located in a dense urban area.
- Precision not 100% ➡ civilian casualties
- A system which needs to identify cancer-risk patients
- Recall not 100% ➡ some patients will die of cancer

Problem of Retrieval

- You give a query q
- You get a ranked list of documents (say 10)
  - $d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8, d_9, d_{10}$
- The document d_i could be "relevant" (+) or "irrelevant"(-)
- Let us assume the relevance's are:
  - +, +, -, -, +, -, +, -, -, +

Precision and Recall

$$Precision = \frac{TP}{(TP + FP)} \qquad Recall = \frac{TP}{(TP + FN)}$$

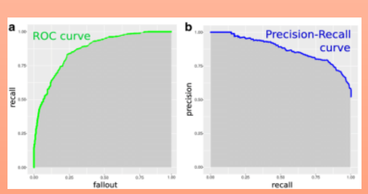
Assume there were **100** "True"/"Relevant" documents (often we do not know this)

	d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8	d_9	d_{10}
Actual	+	+	-	-	+	-	+	-	-	+
Predicted	+	+	+	+	+	+	+	+	+	+
P@K	1.0	1.0	0.66	0.50	0.60	0.50	0.57	0.50	0.44	0.50
R@K	0.01	0.02	0.02	0.02	0.03	0.03	0.04	0.04	0.04	0.05

Classifier Evaluation

Acquire Raw Data → Prepare / Clean Data, Visualization → Feature Engineering → Pick Model and Hyper-params for Task → Model Training / Optimization → Evaluate Model Performance → Deploy Model

- ✓ Performance Evaluation Metrics
  - ✓ Accuracy Metrics
  - ✓ Precision and Recall
  - F-measure (F1 Score)
- ROC and PR curves



PR curves

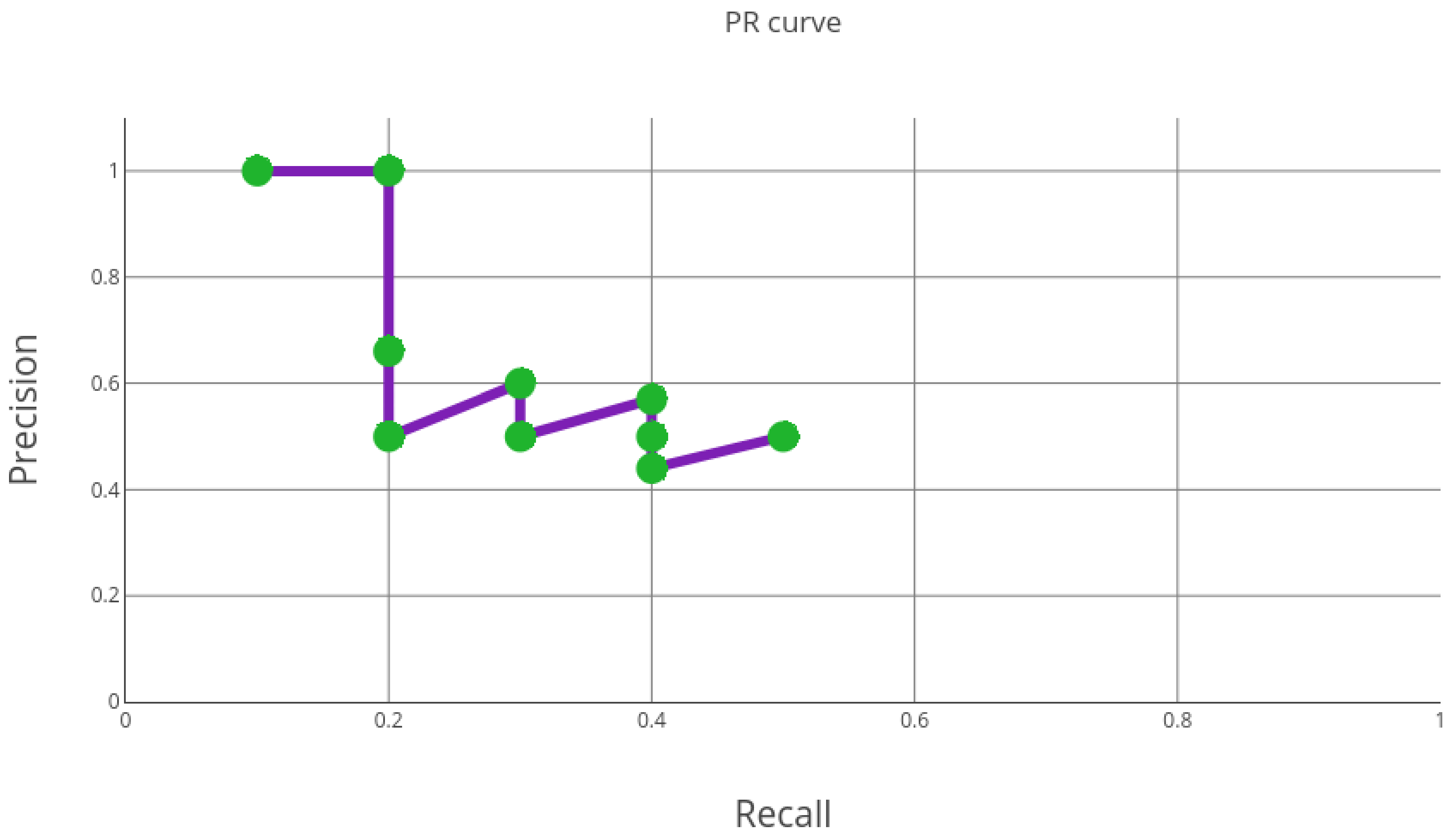
Precision and Recall

$$Precision = \frac{TP}{(TP + FP)} \qquad Recall = \frac{TP}{(TP + FN)}$$

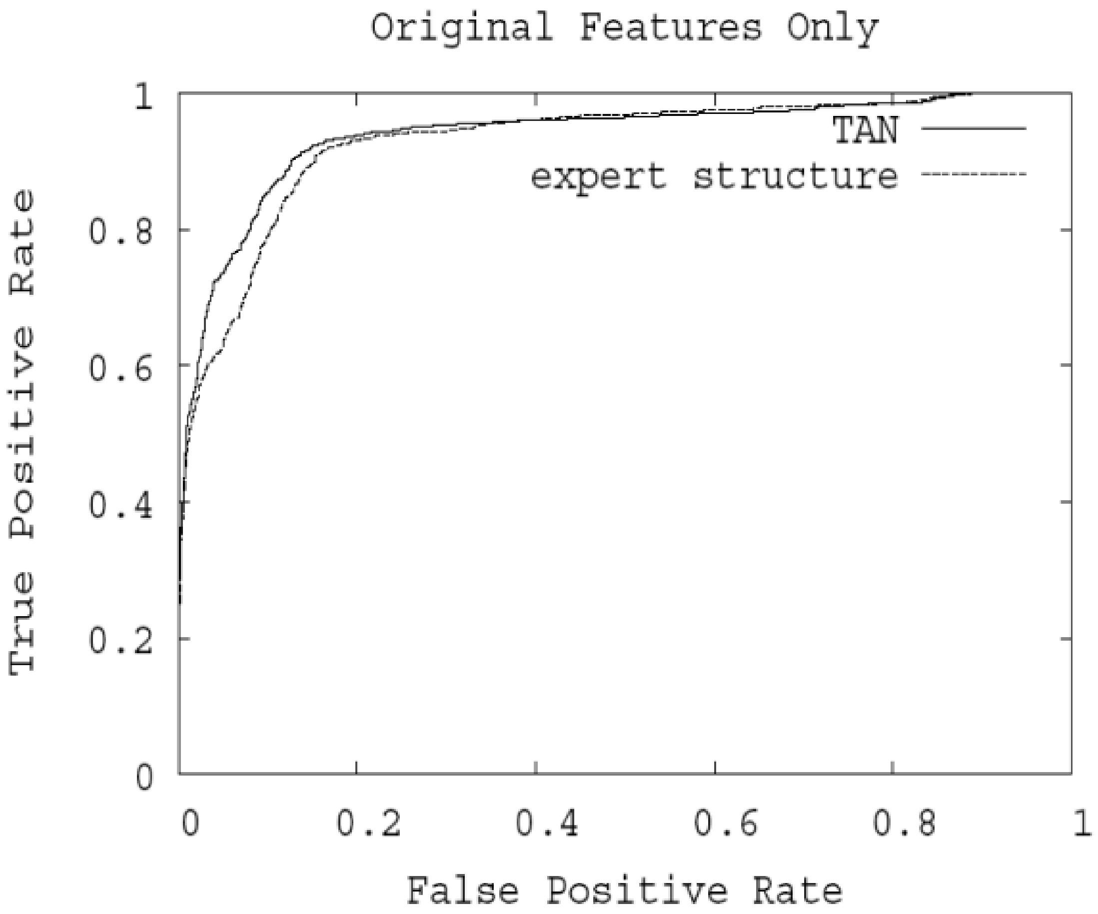
Assume there were **10** "True"/"Relevant" documents (often we do not know this)

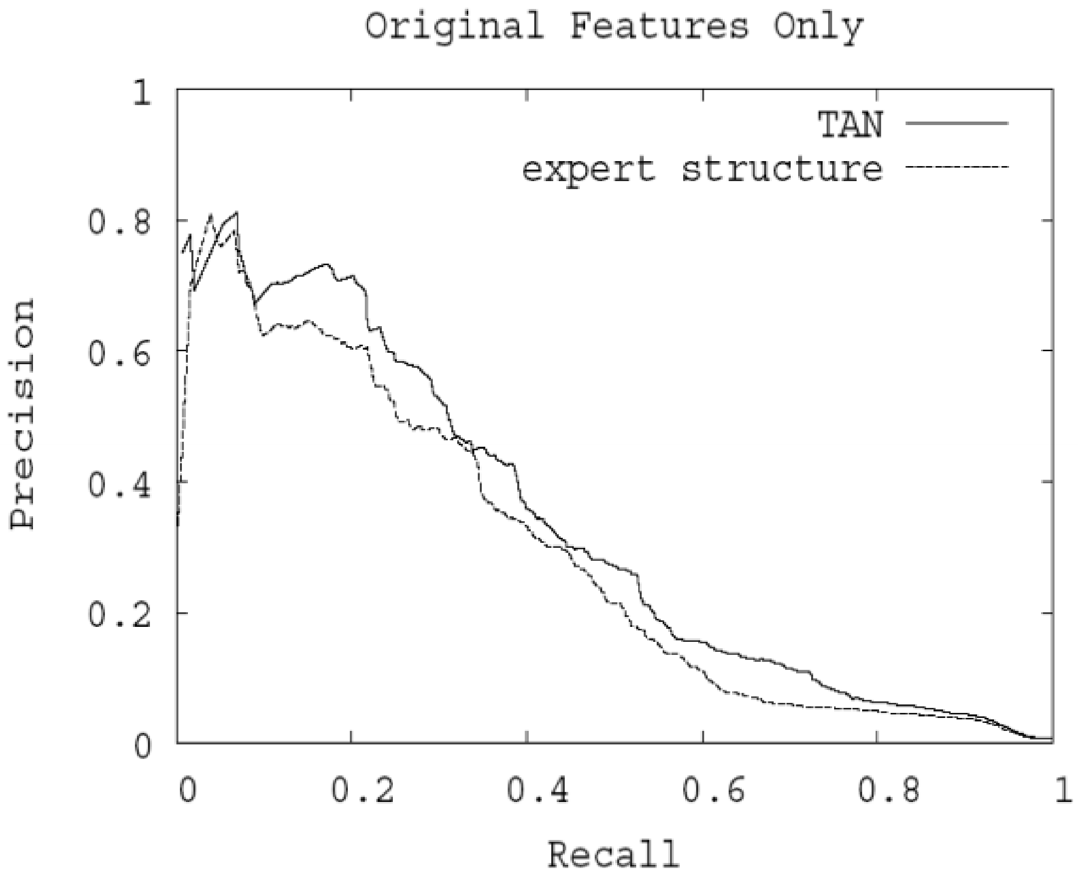
	d_1	d_2	d_3	d_4	d_5	d_6	d_7	d_8	d_9	d_{10}
Actual	+	+	-	-	+	-	+	-	-	+
Predicted	+	+	+	+	+	+	+	+	+	+
P@K	1.0	1.0	0.66	0.50	0.60	0.50	0.57	0.50	0.44	0.50
R@K	0.1	0.2	0.2	0.2	0.3	0.3	0.4	0.4	0.4	0.5

Precision and Recall



ROC + PR Curves Example

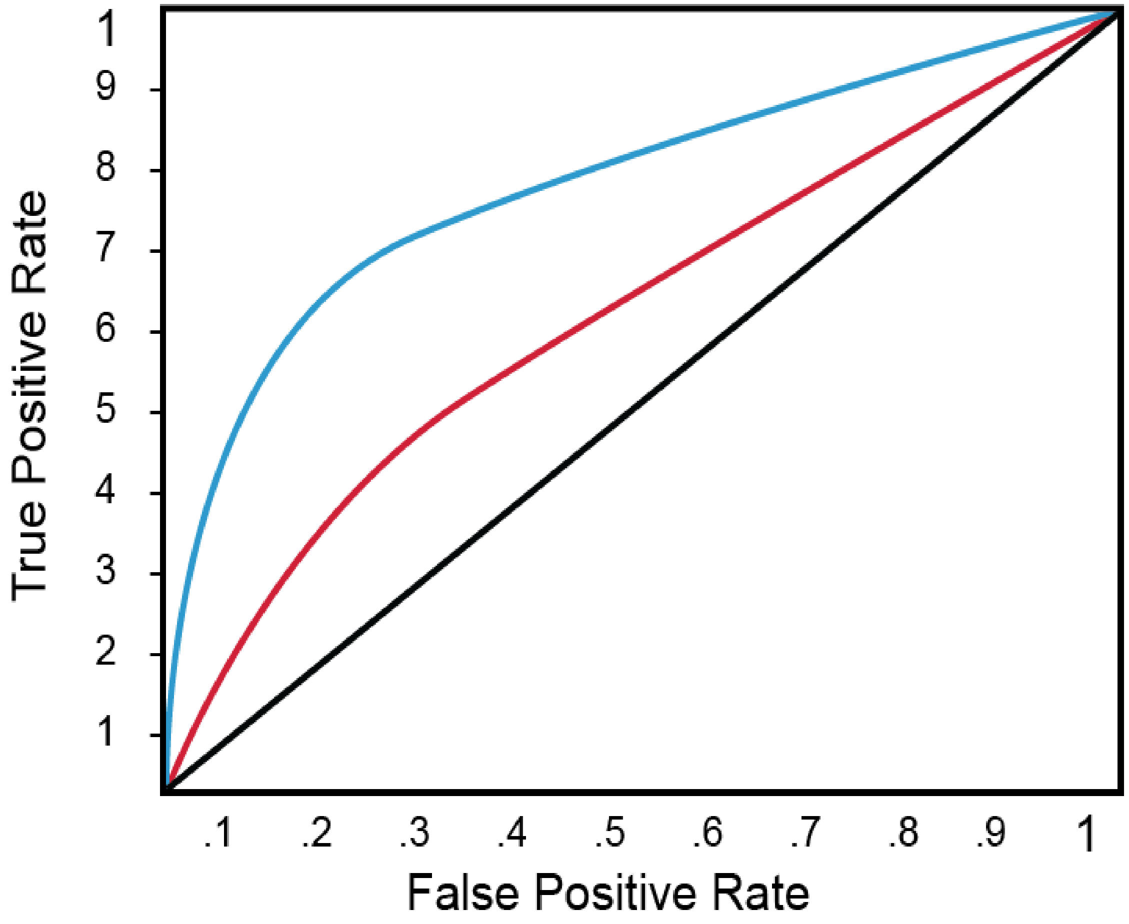




Trade Off...

- To compare two screening tests, at ROC(Receiver Operating Characteristics):

- The higher the Curve, the better.



F-measure: Combines Precision and Recall

- What to do when one classifier has better Precision but worse Recall, while other classifier behaves exactly opposite?
  - F-measure (Information Retrieval)

$$F_1 = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

F-measure

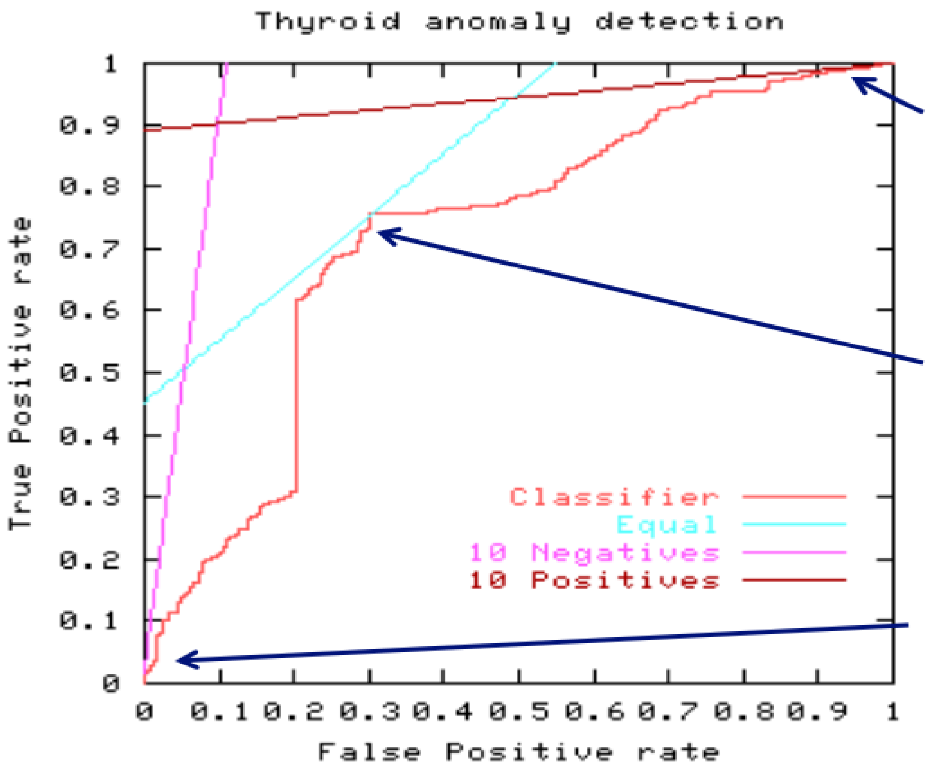
$$Precision = \frac{TP}{(TP + FP)} \qquad Recall = \frac{TP}{(TP + FN)}$$

- What to do when one classifier has better Precision but worse Recall, while other classifier behaves exactly opposite?
  - F-measure (Information Retrieval)

$$F_1 = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

- F1 measure punishes extreme values more !
- Definition of Recall and Precision have same numerator, different denominators. A sensible way to combine them is harmonic mean.

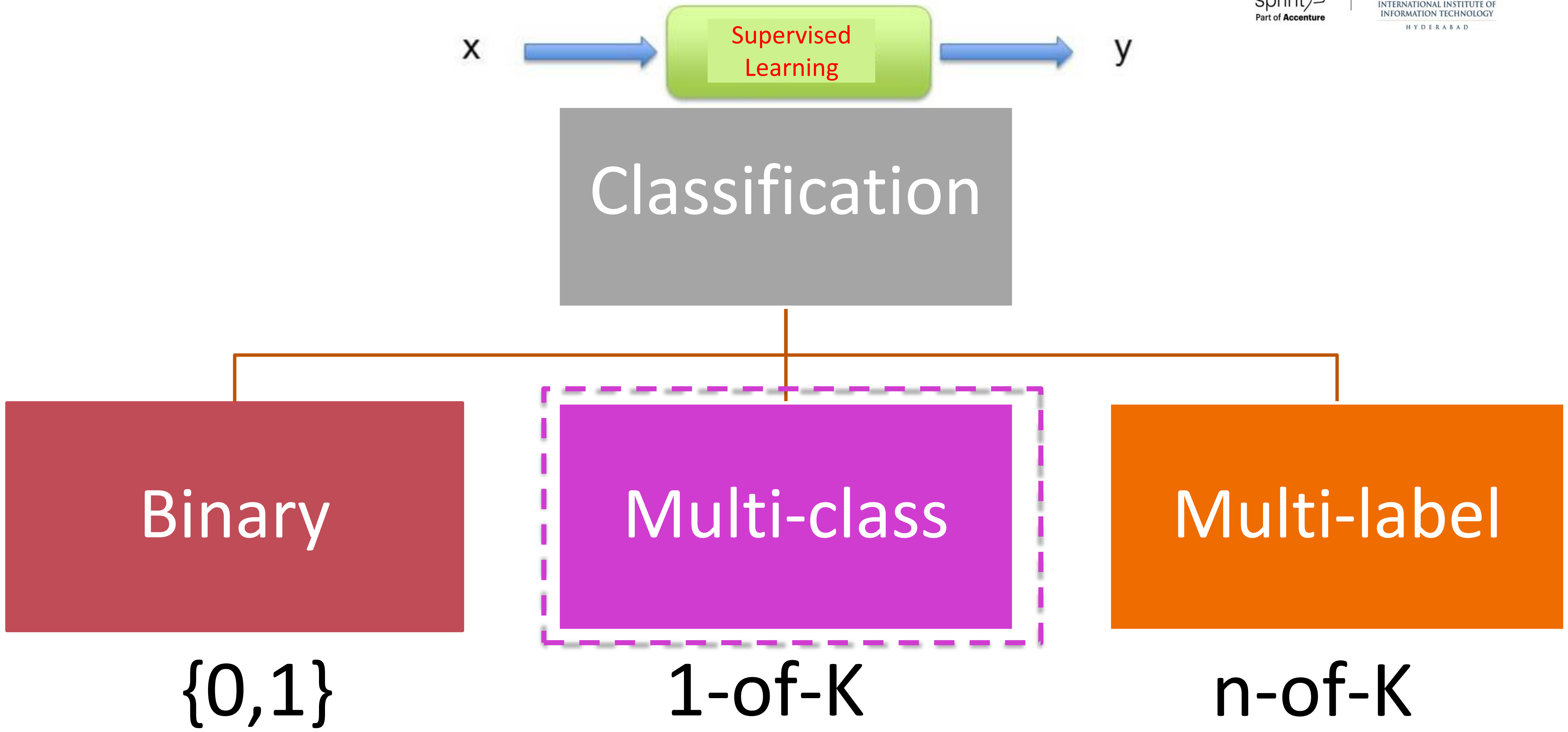
ROC Curves and Misclassification Costs.



best operating point when
FN costs 10× FP

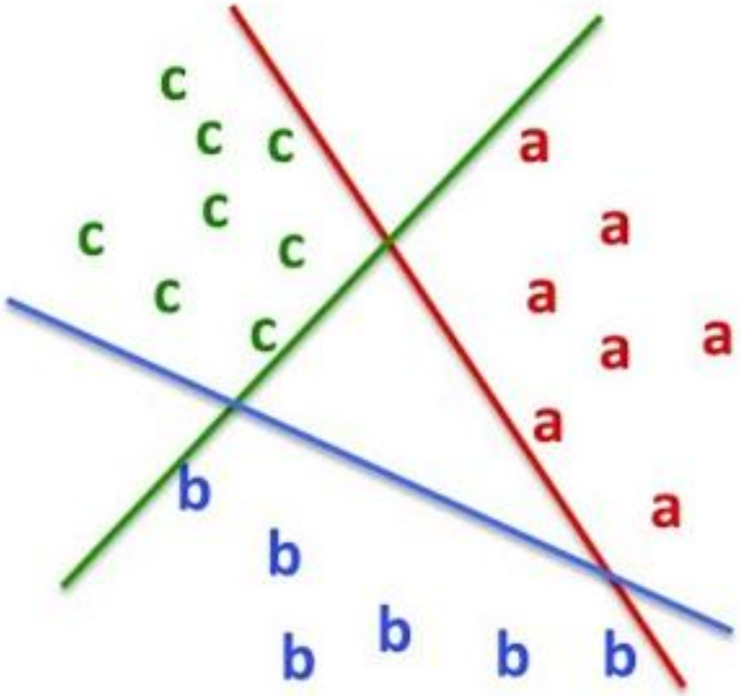
best operating point when
cost of misclassifying positives
and negatives is equal

best operating point when
FP costs 10× FN


x
Supervised Learning
y
Classification
Binary
Multi-class
Multi-label
{0,1}
1-of-K
n-of-K

How to use 2-class measures for multi-class ?

- Convert into 2-class problem(s) !



Multi-class problems - Confusion matrix

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

Avg. accuracy may not be very meaningful with imbalanced class label distribution

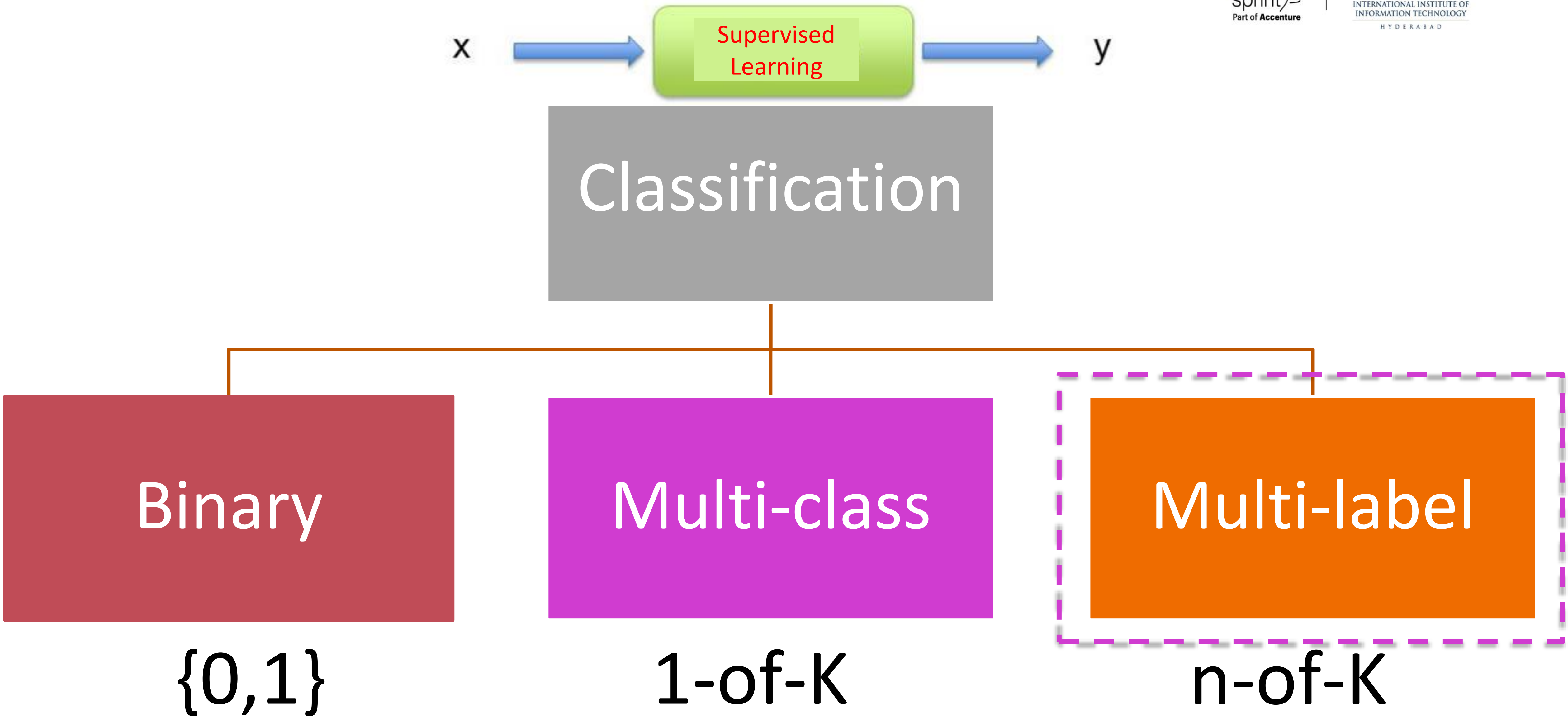
activity recognition from video

actual class

	bend	jack	jump	pjump	run	side	skip	walk	wave1	wave2
bend	100	0	0	0	0	0	0	0	0	0
jack	0	100	0	0	0	0	0	0	0	0
jump	0	0	89	0	0	0	11	0	0	0
pjump	0	0	0	100	0	0	0	0	0	0
run	0	0	0	0	89	0	11	0	0	0
side	0	0	0	0	0	100	0	0	0	0
skip	0	0	0	0	0	0	100	0	0	0
walk	0	0	0	0	0	0	0	100	0	0
wave1	0	0	0	0	0	0	0	0	67	33
wave2	0	0	0	0	0	0	0	0	0	100

predicted class

Courtesy: vision.jhu.edu


x
Supervised Learning
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Classification
Binary
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{0,1}
1-of-K
n-of-K

Summary

- Many metrics:
  - Accuracy, TP, FP, AUC, Precision, Recall, AP/mAP
  - Class imbalance and decision-cost imbalance must be taken into account

- Confusion Matrix: Important to analyse and refine solution.

- Curves provide “Trade off” and help compare classifiers / retrieval systems

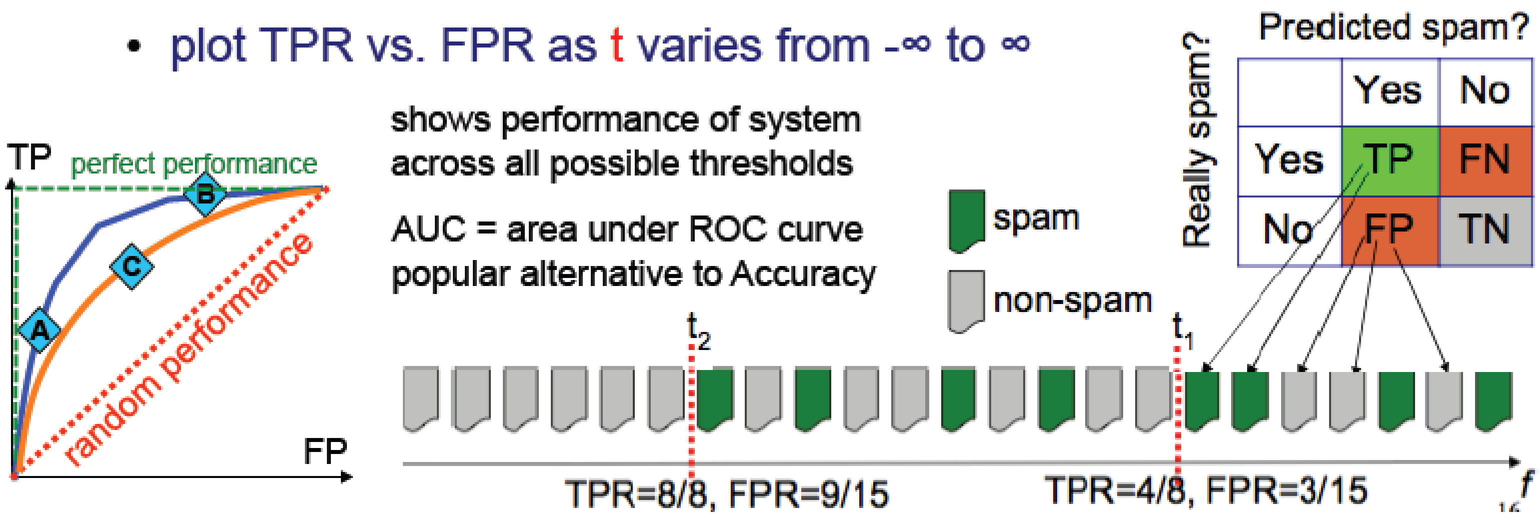
Questions?

Receiver Operating Characteristic

ROC Curves

- Many algorithms compute "confidence" f(x)
  - Threshold to get decision: spam if f(x) > t, non-spam if f(x) <= t
  - Threshold to determines error rates

- Receiver Operating Characteristic (ROC)

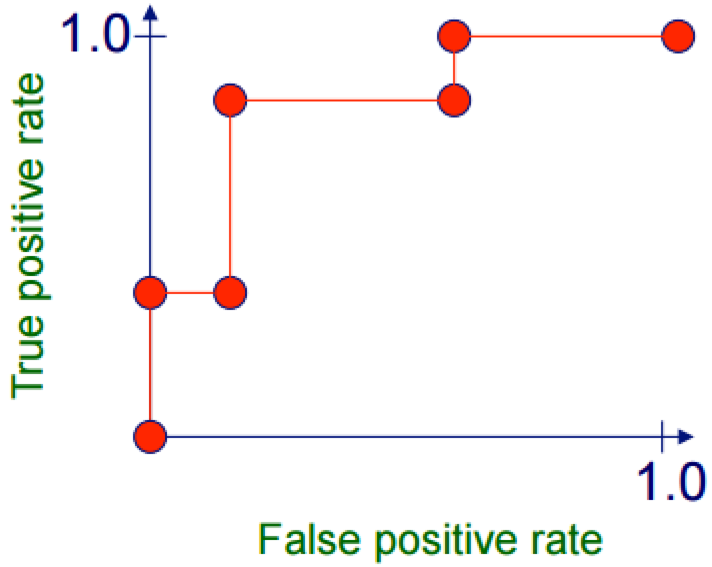


ROC Curve: Algorithm

- Sort test-set predictions according to confidence that each instance is positive
- Step through sorted list from high to low confidence
  - Locate a threshold between instances with opposite classes (keeping instances with the same confidence value on the same side of threshold)
  - Compute TPR, FPR for instances above threshold
  - Output (FPR, TPR) coordinate

Plotting an ROC Curve

instance	confidence positive		correct class
Ex 9	.99		+
Ex 7	.98	TPR= 2/5, FPR= 0/5	+
Ex 1	.72	TPR= 2/5, FPR= 1/5	-
Ex 2	.70		+
Ex 6	.65	TPR= 4/5, FPR= 1/5	+
Ex 10	.51		-
Ex 3	.39	TPR= 4/5, FPR= 3/5	-
Ex 5	.24	TPR= 5/5, FPR= 3/5	+
Ex 4	.11		-
Ex 8	.01	TPR= 5/5, FPR= 5/5	-



Plotting an ROC Curve

- Can interpolate between points to get convex hull

Thanks!!

Questions?